Threshold Forecasting Under Rare Events

Early KPI Forecasting with Survival Curves, Bayesian Updating, and Monte Carlo Simulation Applied to Mini Golf Milestones

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Context

Forecasting performance has never rested solely on statistical precision; it requires disciplined judgment in the face of uncertainty. While most predictive models perform admirably under conditions of data abundance and repetitive outcomes, they often falter when confronted with rare events, limited samples, and compressed decision timelines. The shortcoming lies not only in the scarcity of information but in the averaging effects that obscure the volatility most relevant to real-world decision-making. Within such settings, the operative question shifts away from likelihood in the abstract toward temporal proximity, asking whether an outcome will materialize soon enough to warrant action. Through a deliberately narrow empirical lens, the “Breaking 30” project addresses this problem by observing the Sanicki brothers’ effort to complete eighteen holes of miniature golf in twenty-nine strokes or fewer, playing one round per day across varying courses and seasons in a two-man scramble. This format yields an inherently sparse stream of data, with one observation per day and one binary result per course, serving as a functional analogue for rare-event prediction in commercial and institutional environments. Each season operates as a standalone experiment in cumulative timing, showing how progress toward a defined threshold accrues not linearly, but in episodic, often imperceptible increments. The resulting dataset is modest in size yet saturated with uncertainty, echoing the conditions that render early-stage forecasting both most challenging and most consequential.

Against this backdrop, the central inquiry becomes whether such infrequent breakthroughs can be anticipated with sufficient credibility to influence strategic behavior in advance of their realization. Comparable questions emerge routinely in business: a sales team nearing its quarterly benchmark, a product approaching market readiness, a user segment inching toward churn. In each instance, post hoc recognition nullifies the chance for intervention, while premature action invites misallocation. The analytical imperative is therefore to treat uncertainty not as an obstacle to prediction, but as an operational variable capable of measurement, refinement, and structured communication. In pursuit of that goal, the study employs a triad of probabilistic methods—survival analysis, Bayesian inference, and Monte Carlo simulation—to convert minimal data into incrementally updated probability estimates. Survival functions articulate how the likelihood of success evolves over time; Bayesian techniques incorporate each new round into an evolving belief structure; and simulation techniques render those revised probabilities into decision-ready metrics. Taken together, these tools offer not a remedy for uncertainty but a disciplined approach to contouring its shape, allowing prediction to serve not as a verdict but as a strategic compass calibrated to act before certainty arrives.

Methodology

Analytical design advanced through three ordered stages: first, the construction of a reproducible data foundation; second, the estimation of historical priors to characterize the timing of milestone achievement; and third, the updating of those priors through Bayesian inference and simulation based on live observations. This progression from structure to inference to interpretation demonstrates how limited, sequential data can yield actionable forecasts when uncertainty is treated as an integral component of analysis rather than an afterthought.

Data Foundation

Preparation protocols were designed to enforce both reproducibility and transparency across all stages of ingestion and transformation. Raw Excel score logs from each course and season were incorporated into a standardized data structure and validated through a schema contract that explicitly defined each column's data type, naming convention, and admissible value range. Automated validation checks ensured uniqueness of player-day entries, continuity of daily indices, and alignment between per-hole scores and total-round aggregates. Following validation, all datasets were exported into a structured layer of cleaned CSV files, each accompanied by a series of daily snapshots capturing the active season's evolving state. These snapshots froze the data at every estimation point, thereby enabling full reconstruction of forecast conditions as they existed in real time. By eliminating the ambiguity associated with ad hoc modeling or retroactive revision, this design established a clear provenance for every inference. Although modest in scale, the resulting database—comprising five historical seasons and one live season observed through Day 9—took the form of a right-censored time-to-event panel, where each round constituted a single time increment and sub-30 completion marked the terminal event. Such a structure mirrored the logic of experimental design. Schema governance, chronological integrity, and version control were not procedural niceties but epistemic requirements. In forecasting environments where evidence accumulates slowly, credibility arises less from algorithmic sophistication than from the verifiability of the input stream itself.

Historical Priors

With the data infrastructure established, the analysis turned to the question of how long it typically takes to reach the milestone event, given past performance. Using the cleaned historical dataset, survival functions $S(t)$ were estimated to describe the cumulative probability that a sub-30 round had not yet occurred by day t. The Kaplan–Meier estimator provided an initial nonparametric depiction of timing, capturing a pattern of gradual early progress followed by inflection as players adapted to course conditions. To generalize these trajectories across diverse courses without erasing their specific features, the analysis employed a Bayesian log-normal Accelerated Failure-Time model with hierarchical random effects at the course level. Each course was treated as a partial realization of a broader underlying distribution, enabling partial pooling of information from high-volume to low-volume environments. This framework reduced variance in data-sparse settings while avoiding spurious confidence in idiosyncratic cases.

Posterior inference was drawn from four Markov Chain Monte Carlo chains with 2,000 iterations each. Convergence diagnostics showed R-hat values below 1.01 and effective sample sizes exceeding 1,000 across all parameters. The resulting posterior distribution for pooled completion timing placed the median milestone near Day 27, with a 90 percent credible interval extending approximately from Day 21 to Day 34. Between-course variation ranged from early completions near Day 13 to extended timelines beyond Day 50, underscoring the heterogeneity embedded in course layout, difficulty, and adaptive learning curves. This pooled prior established a probabilistic anchor for subsequent forecasting. Rather than projecting a single day of expected completion, it offered a full distribution of plausible outcomes, allowing uncertainty to enter the model not as residual error but as a quantifiable input.

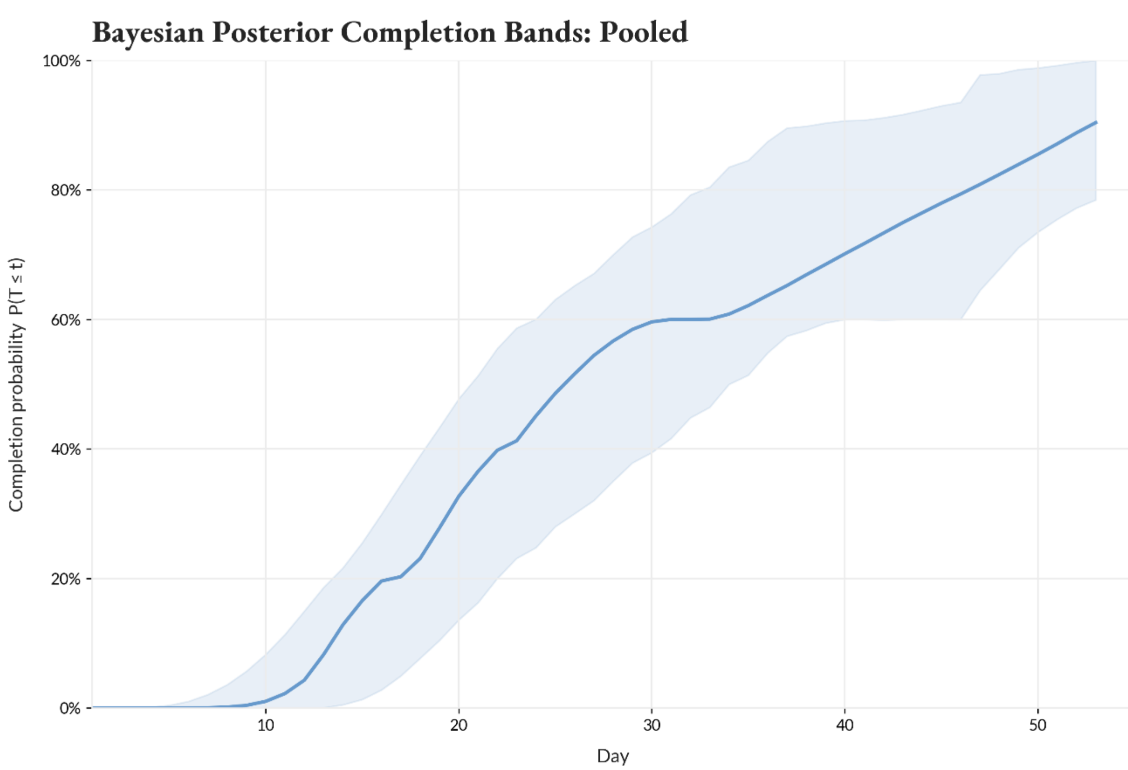

Figure 1. ***Posterior completion probability bands based on pooled historical data.***

The median trajectory indicates that milestone completion typically occurs near Day 27, while the shaded region reflects the 90 percent credible interval, spanning approximately Day 21 to Day 34. The gradual slope and widening uncertainty beyond Day 30 illustrate both the rarity of early success and the persistent possibility of long-tail outcomes. This distribution serves as the empirical benchmark against which live forecasts are subsequently evaluated.

Live Forecast and Posterior Updating

Real-time observations from the ongoing sixth season were incorporated through Day 10 to update the posterior distribution, thereby transforming the historical prior into a live conditional forecast. Each additional round informed the evolving probability structure, progressively narrowing the credible interval as evidence accumulated. After conditioning on observed performance, the updated median forecast shifted toward Day 29, with probability mass concentrated in the late-month window between Days 25 and 30. Compared to the historical prior, this posterior showed reduced variance, reflecting a consistent pattern of mid-30s rounds that constrained the plausible range for early breakthroughs.

To render these posterior shifts more interpretable for strategic purposes, the model generated 10,000 Monte Carlo simulations to represent plausible futures under current conditions. Each simulated trajectory drew from empirically estimated hole-level probability mass functions, calibrated using recent and historical data. These simulations did not replace the time-to-event model but rather translated its implications into outcome distributions, converting probabilistic structure into intuitive operational metrics. Model diagnostics confirmed internal coherence. Posterior trace stability, posterior predictive overlays, and alignment between observed and expected hole outcomes verified convergence and calibration. The live posterior, shown in Figure 2, illustrates how early-season data re-centered and tightened the expected completion window, translating statistical uncertainty into a credible, decision-ready probability surface.

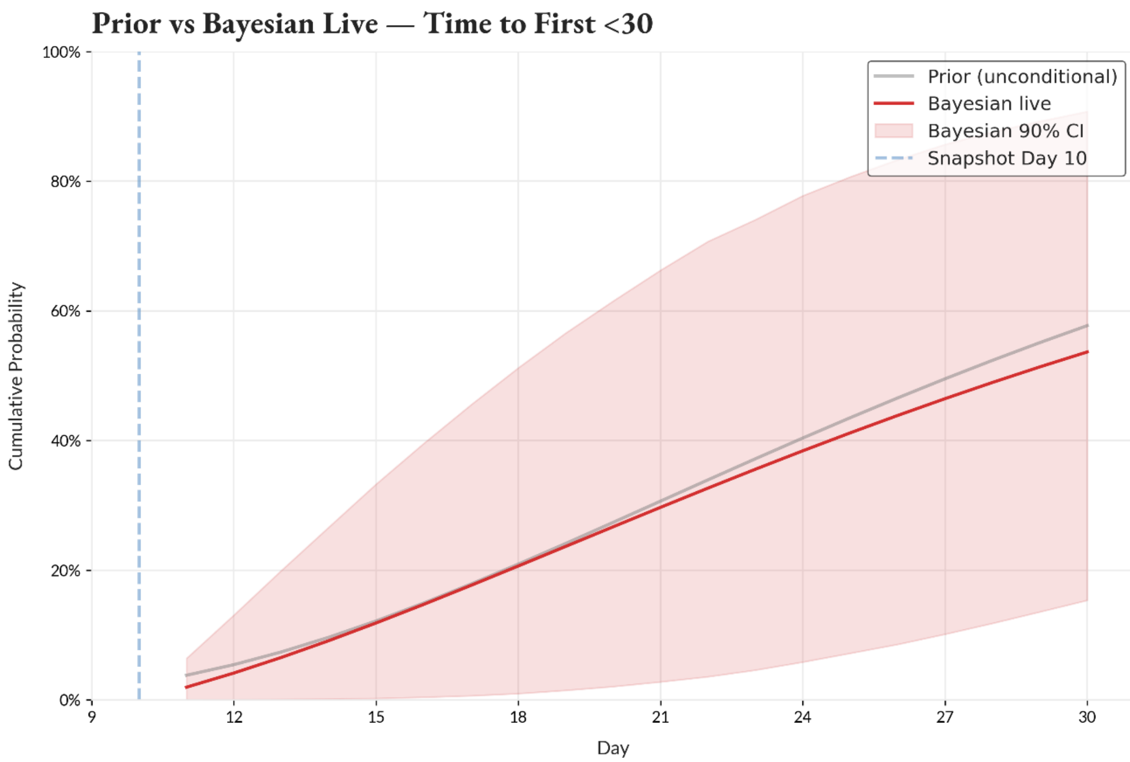

Figure 2. Bayesian forecast after ten days (red) compared with the pooled historical prior (gray).

Incorporating early rounds narrows the credible interval and shifts the median completion estimate to Day 29, delineating a plausible window for a successful sub-30 round while quantifying the risk of delay.

Findings

The model's results illuminate two dimensions of the forecasting problem: first, the inherent variability in historical milestone timing; and second, the extent to which live observations can substantively constrain that uncertainty. Among the five completed seasons, milestone completions ranged widely from Day 13 to Day 52, with the pooled median settling near Day 26. This level of dispersion underscores a structural challenge in rare-event forecasting. Early-stage signals of improvement tend to be obscured by ambient variance, rendering breakthroughs difficult to anticipate until performance patterns begin to stabilize across repeated trials. Incorporating ten days of live data materially reduced that ambiguity. Relative to the pooled historical prior, the posterior credible interval surrounding expected completion contracted by approximately half, with probability mass now concentrated between Days 25 and 30 and a median near Day 29. Such narrowing of the predictive band illustrates that Bayesian updating does more than adjust forecasts in response to new evidence. It strengthens the model's inferential structure by converting a diffuse projection into a concentrated probability surface. In effect, the model inferred from consistent mid-30s rounds, despite the absence of a sub-30 outcome, that proximity to the milestone had become statistically credible.

A parallel perspective emerges through Monte Carlo simulation. Across 10,000 simulated rounds, the distribution of outcomes was sharply centered around 33 to 34 strokes, with only 0.93 percent of paths achieving the sub-30 threshold. This narrow mode signals a high degree of repeatability in current performance, yet also reveals that such repeatability has not yet translated into the efficiency required to clear the milestone. Importantly, simulation variance declined in tandem with the empirical stability of observed scores, reflecting a pattern of convergence rather than randomness. What emerges is not volatility but constraint: the limiting factor is not erratic deviation but the infrequency of high-leverage gains. Even modest improvements in early-hole conversions or stroke precision would recalibrate the entire probability distribution. This underlying learning dynamic is reflected in the evolution of the model's probability curve. As early rounds accumulated, the lower bound of expected performance rose, the left tail of the distribution flattened, and the central mass tightened. Rather than producing a late-stage spike, the model registered a gradual increase in completion likelihood, signaling the steady accumulation of information rather than a sudden regime shift. In statistical terms, consistency became the driver of clarity. As variability declined, the model's posterior sharpened, revealing not just a likely outcome but a plausible mid-season window for milestone achievement.

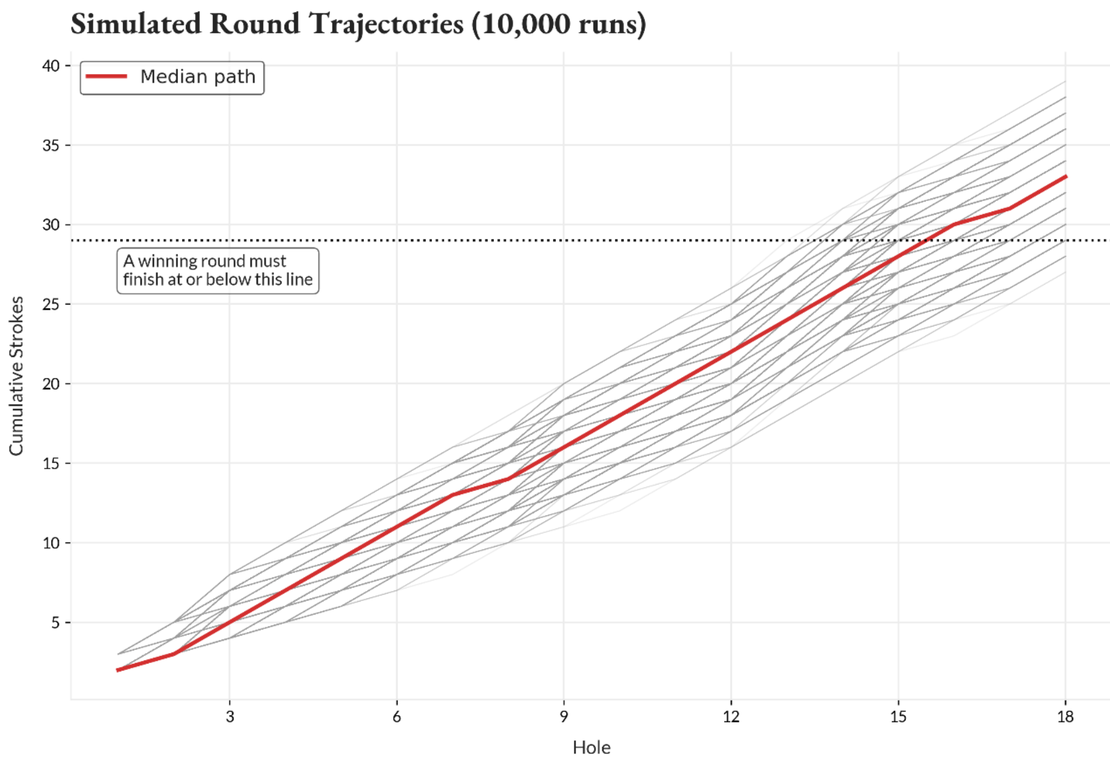

Figure 3. Monte Carlo simulation of 10,000 possible rounds.
Most simulated outcomes cluster around 33 to 34 strokes, with only 0.93 percent meeting the sub-30 threshold, quantifying the narrowness of the current path to success.

Implications

By repositioning uncertainty as a generative input rather than a forecasting obstacle, the findings clarify how probabilistic methods offer more than statistical abstraction. Within this framework, incomplete information is not treated as a defect to be corrected but as the epistemic foundation upon which credible inference is constructed. The model does not aim to eliminate ambiguity; it quantifies it directly by assigning probabilities to both the timing and magnitude of events. What results is not a fixed-point forecast but a dynamic distribution—an evolving structure of belief that sharpens incrementally with each new observation.

Structured in this way, the modeling approach holds immediate relevance for domains far removed from its empirical origin. Conditions such as sparse measurement, delayed feedback, and forward-looking decision pressure are common to commercial forecasting, policy planning, and operational risk management alike. Whether estimating the likelihood of quarterly revenue attainment, gauging product adoption trajectories, or monitoring thresholds for regulatory intervention, the underlying problem remains consistent: decisions must be made before certainty arrives. Under such conditions, Bayesian updating offers not a guarantee but a discipline. It allows prior expectations to adapt systematically as evidence accrues, while maintaining transparency about the assumptions that remain unsettled. Equally foundational is the question of interpretability. Probabilistic outputs must be more than technically valid; they must also be communicable across institutional contexts. Credible intervals and posterior probabilities provide that bridge. A projection stating a seventy percent chance of milestone completion by a specific date enables scenario planning, guides resource deployment, and renders the forecasting process intelligible to those not trained in statistical inference. In this sense, clarity does not mean simplification. It means structural transparency that makes belief accountable.

Figures 1 through 3 collectively map this transition from statistical abstraction to operational insight. The historical priors illustrate that rare outcomes can still produce informative distributions. The live posterior shows that even in the absence of success, consistent intermediate performance reduces uncertainty meaningfully. The simulation layer translates variance into practical probability surfaces, showing where small improvements would carry the greatest marginal effect. Each layer contributes not only to the predictive arc but to the broader institutional function of the forecast: to inform action before the event is settled. What emerges is a principle that generalizes. Success in conditions of uncertainty cannot depend on the removal of variability. It must depend on measuring that variability clearly enough to support timely, accountable action. Probabilistic forecasting, when constructed as an evolving system of belief, does not promise perfect foresight. It furnishes the structure within which future uncertainty becomes navigable.

Reflection

While the analysis presents a coherent structure for forecasting rare milestone events, its empirical foundation remains intentionally constrained. Each season contributes only a single terminal outcome—whether or not the milestone is achieved—which limits statistical power and restricts visibility into repeated learning dynamics within the same environment. The log-normal distribution assumed in the accelerated failure-time model offers interpretive tractability, yet may mask underlying complexities if the hazard function is shaped by nonlinear or multimodal forces. In addition, the model treats daily performance as conditionally independent, abstracting away from plausible autocorrelations in skill, confidence, or external conditions. These limitations suggest clear paths for extension. Future work could enrich the dataset by incorporating covariates that proxy for course complexity, environmental variation, or player fatigue. Conditional hazard estimation would then reflect behavioral and situational nuance rather than rely on homogenous timing assumptions. A shift to a continuous-time framework, rather than indexing by discrete days, could also sharpen temporal sensitivity, particularly in capturing the rhythm of cumulative skill acquisition or decay. Moreover, longitudinal performance indicators—such as putting conversion rates, first-attempt accuracy, or mean proximity to the cup—could support hierarchical modeling of per-hole improvement across seasons, linking micro-level trends to macro-level outcome probabilities.

Even in this narrower form, however, the framework lends itself to broader application. Operational forecasters in business contexts can adapt survival models to anticipate when revenue thresholds, usage rates, or cost targets are likely to be met. Policymakers can model the diffusion of regulatory impacts or benefit uptake with similar tools. Process engineers and strategic planners can use analogous techniques to detect inflection points where intervention carries disproportionate value. Across these contexts, the analytical posture remains consistent: forecasting becomes viable not through removal of uncertainty, but through its explicit measurement. At its core, the model affirms that disciplined probabilistic reasoning allows rare-event forecasting to move beyond conjecture. By embedding belief revision within a transparent structure, the analysis demonstrates how even sparse data can support inference without overstating precision. The value lies not in producing definitive forecasts, but in constructing conditional expectations that evolve intelligibly. Treating uncertainty as a measurable property rather than a deficiency permits the analyst to engage with reality as it unfolds, rather than retroactively explain it. This orientation, more than any particular technique, is what renders early forecasting actionable.